

Tushar Gadkari, PhD

245 Marion Oaks Ln, Ocala, FL 34473 | tg@belaporeanalytics.com | (786) 218-2293

EXECUTIVE SUMMARY

- Data science leader with successful experience designing and driving data modeling initiatives to improve financial performance and achieve business goals.
- Strong experience working with C-Suite executives to assist in strategic decision making, translate company goals into data strategies, transform operations, and increase profitability. Proven manager with ability to identify, lead, and train teams of data scientists, quantitative developers, and third-party vendors, both on-premises and offshore.
- Expert in data analytics and machine learning. Led development of multiple predictive credit risk, risk-based pricing models for loan decisioning, and marketing lead-gen models, customer lifetime value (CLV), customer churn models to improve customer acquisition and retention.
- Proficient in time-series, forecasting, A/B experimentation and conducting market and customer segmentations. Tests hypotheses and builds statistical models to evaluate and implement proposed business strategies and deployment of real-time feedback responses.
- Leverages advanced competencies with programming, including R, PYTHON, MATLAB, Shell scripting, SQL, AWS, UNIX, to achieve superior productivity. Drives cloud-focused, data-centric infrastructure initiatives.
- Technical author with experience publishing and presenting key data in peer-reviewed journals, conferences, blogs for business, finance, and strategic purposes.

WORK EXPERIENCE

BELAPORE ANALYTICS Ocala, FL, USA

October 2022 to Present

Principal Consultant, www.belaporeanalytics.com

- Strategic executive focused on providing data analytics and machine learning services to financial services companies. Spearheaded implementation of multiple advanced analytics initiatives. Demonstrated success in introducing cutting-edge machine learning algorithms to boost performance. Influences decision making at the executive level by translating complex data into digestible insights.
- Expert at building and deploying predictive models to optimize business processes, implement data-driven risk management, and forecast key performance indicators. Revamps data strategy to improve visibility of key performance metrics. Creates interactive dashboards that provide real-time insights into revenue and operations.
- Skilled in applying advanced modeling techniques, enabling clients to anticipate market trends, customer behaviors, or operational challenges. Improves customer lifetime value and retention via targeted profiling. Focuses on achieving business goals by implementing data mining techniques to increase marketing campaign effectiveness and reduce customer churn.
- Partners with cross functional teams to align data initiatives and enable data-driven decision making. Champions operational efficiency through process optimization. Adept at leveraging big data tools for extracting meaningful insights from disparate and large data sets. Builds data pipelines, implements AWS microservices for production model deployments.
- Utilizes Python, R, SQL to perform statistical analysis and drive machine learning algorithm initiatives.

ANNIE MAC HOME MORTGAGE Mount Laurel, NJ, USA

July 2020 to September 2022

Vice President, Data Science

- Onboarded by the executive team to improve quality of mortgage and real estate leads with novel data solutions. Conceptualized HomeHunt.com - a data driven web portal for digital leads acquisition. Primary objective of the HomeHunt initiative was to drive the consumer forward through the process pipeline while improving top of the funnel engagement and optimizing for the real estate and mortgage transaction cost. Facilitated spin-off of HomeHunt.com as an independent sister entity of AnnieMac Home Mortgage.
- Efforts led to in-house leads generated with cost per lead (CPL) of \$21 compared to \$55 CPL for purchased Zillow leads of equivalent quality and conversion. Achieved a 6% lead transfer rate from HomeHunt's call center to AnnieMac loan officers, results consistent with purchased Zillow's long-form lead transfer rate of around 5-10%.
- Built a core data science team and recruited critical hires and consultants for HomeHunt digital marketing team. Conducted project management, vendor selection and led data decisions. Designed data architecture and led the development of data infrastructure on AWS to store consumer profiles and third-party customer datasets. Worked closely with technology team to drive AWS data migration and facilitated cloud adoption. Drove efforts to sanitize and streamline CRM data along with identifying high impact process changes through data mining. Successfully implemented Amperity customer data platform (CDP) to obtain 360° holistic view of the customer journey and to store proprietary property and mortgage data assets.

- Identified and acquired datasets from Jornaya, Infutor, V12, and Corelogic to enrich lead profile information such as identity verification, identity resolution, and identity profiling to supplement first party data. Obtained third party online activity datasets including In-Market Pre-Mover activity, Online browsing activity, Shopping events and triggers and Time window of web behavior to build customer profiles and audience segmentations. Led development of APIs to fetch third-party datasets.
- Developed and deployed lead-gen models to identify high propensity leads, built classification for good/bad leads using an ensemble of clustering models, and a logistic model to produce a custom scorecard for likelihood of lead conversion and customer churn. Identified attributes correlating with lead performance and built customer lifetime value (CLV) metrics from first-party internal data and third-party data to increase customer retention. Assisted with scaling on ad spend by optimizing on lead conversions and CPL.
- Played an active role in strategizing digital marketing initiatives and process development. Designed UTMs for lead tagging and tracking across multichannel marketing campaigns. Conducted IVR and call log analysis on ring central data from the HomeHunt call center. Built audience segmentations for a list of targeted campaigns for home buyer/seller, first time home buyer/seller and multiple nurture campaigns. Participated in IDX implementation discussion to drive strategy with consultants. Tested A/B scenarios on IDX layouts and customer responses to deliver search and content experience. Facilitated rollout in five cities to test customer acquisition and experience of personalized customer touch along with obtaining first party customer persona data.
- Aggregated and delivered real-time insights for multichannel digital marketing initiatives and provided business process recommendations from data insights. Led the development of Power BI dashboards and designed the underlying SQL tables to track lead funnel, ad spending, and customer KPIs. Conducted ad-hoc data analytics using R, Python and SQL. Provided automated weekly KPI insights (Monday memos) to the executive team followed with a detailed monthly presentations and quarterly onsite project progress updates and blueprints for upcoming quarter's plan of action.
- With a focus on data security and regulatory best practices, developed a scalable, compliant system that allowed HomeHunt to grow rapidly and operate within the national framework of overlapping MLS coverage areas allowing access to consumer data and monetizing resultant leads with AnnieMac.

BANKERS HEALTHCARE GROUP Davie, FL, USA

April 2019 to May 2020

Senior Manager, Revenue Analytics

- Recruited by company COO to lead the development of subprime loan portfolio function. Built model-driven credit risk and pricing strategy to incorporate targeted origination growth and expected loan defaults. Collaborated with the Chief Risk and Chief Revenue Officer to conceptualize the product blueprint and strategize implementation.
- Designed and led the development of sub-prime loan offerings to allow credit approval to higher-risk population, while controlling risk exposure. Implemented A/B experiments for acquisition of credit model and pricing strategies into operations.
- Introduced direct-mail marketing campaign for subprime population, built classifiers, and created segmentation within data to identify target population, and improve process operational efficiency, reduce customer churn through data-driven automation efforts. Efforts led to \$5M increase in monthly loan originations.
- Drove efforts to improve data architecture. Led deployment of predictive algorithms and mathematical models into production with introduction of Machine Learning deployment platforms, like deployR. Led API efforts to bring third-party data, including bank data from Yodlee, for process automation and improvement of data quality.
- Leveraged Data Analytics using R, Shiny, MySQL, Python and Linux bash scripts to implement methodologies for real-time tracking of subprime portfolio delinquencies and risk-model KPIs.

BFS CAPITAL Coral Springs, FL, USA

August 2015 to March 2019

Predictive Analytics Lead, Risk

- First employee hired by President and CEO to create and lead a data science group. Reported directly to the CEO and interacted daily with all tiers of management, including underwriters, members of the credit committee, marketing, and sales. Conducted comprehensive Data Analytics, provided business insights with interactive KPIs, and made regular presentations to C- level executives and the Board.
- Developed credit risk, fraud, and bankruptcy models to minimize defaults in BFS Capital's \$300M commercial loan portfolio, improved portfolio profitability by reducing losses in the small ticket loans from 21% to sub 10%. Built risk-based loan pricing models, structured paper grades for loan credit quality, and implemented pricing strategies to improve portfolio yield and performance.
- Managed API integration projects to create data pipes and facilitate transition to a cloud-centric data-driven Machine Learning infrastructure on AWS. Integrated big datasets from credit bureaus and third-party resources to produce a borrower's risk profile, built classifiers, conducted feature engineering, and implemented Machine Learning techniques such as Random Forest, Gradient Boosting Machines to identify attributes of interest for customer profiling, designed credit score cards for preapproval of retail small business loan borrowers.

- Led efforts to streamline and automate operations with implementation of prequalification scoring processes that allowed the company to improve speed of service and customer satisfaction while improving performance of credit segment. Worked closely with marketing team on outreach strategies, including lead volume, conversion rates, and optimization of customer acquisition costs.
- Built and managed a high-performing data science team, including PhD level analysts, quantitative developers, and offshore IT vendor resources, led the professional development and productivity of the team. Implemented interactive dashboards to continuously monitor portfolio performance, aging, and delinquencies. Supported new product development via analysis and mathematical models.

OCEAN RIDGE BIOSCIENCES West Palm Beach, FL, USA

June 2014 to July 2015

Data Scientist, Genomics

- Built proprietary intellectual property of genomic databases (ensembl, mirExplorer, miRbase) to query for miRNAs mapping to genetic sequences, leading to a creation of new business line for the company. Efforts led to an increase in sales of genomic data licensing, mapping technology, and related services to clients, facilitating 15% annual additional revenue for the company.
- Led statistical research and technological innovation. Improved operational efficiency by upgrading older implementations of open-source programs and spreadsheet calculations with in-house statistical analysis programs developed in R. Implemented MySQL routines, Python, and bash scripts used to build automation pipelines for DNA sequencing data analysis. Conducted comprehensive statistical analysis to test the elevation of gene expression in disease/treatment scenarios for client projects.
- Drove automation efforts and minimized process downtime by implementing Twiki-based intranet portal (mvc framework) to enable customer project tracking and inventory database management. Improved average project turnover time from 5 weeks to 14 days.

FLORIDA INTERNATIONAL UNIVERSITY Miami, FL, USA

August 2007 to May 2014

Graduate Research Assistant, Vascular Physiology and Biotransport Lab

- Developed nonlinear mathematical models to validate empirical observations of endothelial oscillatory Ca^{2+} -frequency-dependent microvascular tone regulation. Implemented ad-hoc Newton Raphson optimization algorithms to determine elasticity parameters of rat mesenteric arterioles. Performed statistical analysis, regression, and hypothesis testing on acquired data using MATLAB. Published research in esteemed peer-reviewed journals and conference proceedings.
- Awarded \$1.5M in NIH research grant, efficiently managed funds, and prioritized resources towards critical and time sensitive projects. Met timely IACUC regulatory requirements and deadlines, proposed necessary policy changes for maintaining IACUC compliance. Received prestigious Dissertation Year Fellowship.
- Managed and mentored undergraduate research students, created documentation of lecture material, and conducted tutorials and workshops for R, MATLAB, and COMSOL software applications.

MCMaster UNIVERSITY Hamilton, ON, Canada

August 2005 to August 2007

Graduate Research Assistant, Signal Processing and Biosimulations Lab

- Developed finite element and Monte Carlo simulations to model Navier-Stokes equations and stress-strain relationships describing the hemodynamics of arterial wall. Built Stochastic models using COMSOL and MATLAB.
- Implemented inverse statistical modeling techniques to predict atherosclerotic regions in the artery from simulated noisy blood flow measurements. Developed numerical methods and optimization techniques to estimate the cerebral blood flow using electromechanical coupling and impedance tomography measurements.

EDUCATION & PROFESSIONAL DEVELOPMENT

2013	FLORIDA INTERNATIONAL UNIVERSITY <i>Doctor of Philosophy (Ph.D), Biomedical Engineering</i>	Miami, USA
2007	MCMaster UNIVERSITY <i>Master of Applied Science (M.A.Sc), Electrical & Computer Engineering</i>	Hamilton, Canada
2004	MUMBAI UNIVERSITY <i>Bachelor of Engineering (B.E), Instrumentation Engineering</i>	Mumbai, India

SOFTWARE EXPERTISE

Programming: R, PYTHON, MATLAB, COMSOL, Shell Scripting

Database: SQL, NoSQL (MongoDB)

Productivity: AWS, UNIX, GIT, Emacs, LaTeX